Magnetic fields

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Interpret magnetic field patterns and compass direction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 9.*
5 A in each branch.

02. Worked example: explain the model
Arrows leave N and enter S outside; compass arrows tangent to field.

03. Read and interpret the model
External arrows N to S; compass north end tangent to the local field.

04. Apply the physics
N-N repels; N-S attracts.

05. Explain your reasoning
A nonzero field at one point has one net direction; crossing lines assign two different directions.

06. Review: connect an earlier idea *Review: Unit 6, Day 18.*
Bends toward the normal; frequency stays the same at a stationary boundary.

07. Exit check
A stronger field in a consistently drawn field map.

08. STRETCH: test the model
Each ordinary piece has both N and S; cutting does not isolate a single pole.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 9.*
Parallel 3 Ω; total 8 Ω; supply current 2 A.

02. Worked example: explain the model
Arrows leave N and enter S outside; compass arrows tangent to field.

03. Read and interpret the model
External arrows N to S; compass north end tangent to the local field.

04. Apply the physics
Opposing fields may nearly cancel; Earth's field, friction and measurement sensitivity matter, so it does not prove all fields vanish.

05. Explain your reasoning
A nonzero field at one point has one net direction; crossing lines assign two different directions.

06. Review: connect an earlier idea *Review: Unit 6, Day 18.*
Thermal peak becomes 1/4 by Wien's law; sound wavelength becomes 1/6 by $\lambda = v/f$. Different models and assumptions.

07. Exit check
No; each piece has north and south poles.

08. STRETCH: test the model
Like poles repel with field bending away from the central region; N facing S attracts with connecting field lines. Qualitative superposition, no lines crossing.